

ESEMPIO N.3

SUBJECT: TWISTRONICS

TITOLO PAPER: “SU(4) Heisenberg model on the honeycomb lattice with exchange-frustrated perturbations: Implications for twistronics and Mott insulators”

query con cui ho interrogato inspire hep:

<https://inspirehep.net/api/literature?q=t%20twistronics%20and%20date%202016-%202026&sort=mostcited>

OUTPUT INSPIRE:

(Considero solo la entry del paper preso in considerazione), il paper scelto ha poche figure apposta per non sprecare api call.

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in connection to correlated insulators such as the eg Mott insulator Ba3CuSb2O9 and the
insulating phase of magic-angle twisted bilayer graphene at quarter filling. Here we
provide a unified discussion of these systems by investigating an extended model that
includes the effects of Hund's coupling and anisotropic, orbital-dependent exchange
interactions. Using a combination of mean-field theory, linear flavor-wave theory, and
variational Monte Carlo, we show that this model harbors a quantum spin-orbital liquid
over a wide parameter regime around the SU(4)-symmetric point. For large Hund's
coupling, a ferromagnetic antiferro-orbital ordered state appears, while a valence-bond
crystal combined with a vortex orbital state is stabilized by dominant orbital-dependent
exchange interactions."
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in connection to correlated insulators such as the  $\text{e}_g$  Mott insulator
 $\text{Ba}_3\text{CuSb}_2\text{O}_9$  and the insulating phase of magic-angle twisted bilayer
graphene at quarter filling. Here we provide a unified discussion of these systems by
investigating an extended model that includes the effects of Hund's coupling and
anisotropic, orbital-dependent exchange interactions. Using a combination of mean-field
theory, linear flavor-wave theory, and variational Monte Carlo, we show that this model
harbors a quantum spin-orbital liquid over a wide parameter regime around the SU(4)-
symmetric point. For large Hund's coupling, a ferromagnetic antiferro-orbital ordered
state appears, while a valence-bond crystal combined with a vortex orbital state is
stabilized by dominant orbital-dependent exchange interactions."
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exchange-frustrated perturbations: implications for twistronics and Mott insulators" }
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-----ANALISI PAPER SENZA PIPELINE VISIVA-----

ANALISI PDF:

comando: python main.py analyze --pdf downloaded_pdfs/"1751397_SU(4) Heisenberg model on the honeycomb lattice wi.pdf" --model gemini

OUTPUT:

Output JSON:

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{
  "problem_addressed": "The paper investigates the effects of exchange-frustrated and Hund's
coupling induced interactions on the SU(4) Heisenberg model on the honeycomb lattice, providing a
unified discussion of correlated insulators such as BaCuSbO and the insulating
phase of magic-angle twisted bilayer graphene at quarter filling.",
  "main_contributions": [
    "Proposes a unified Kugel-Khomskii model incorporating Hund's coupling and anisotropic,
orbital-dependent exchange interactions on the honeycomb lattice",
    "Demonstrates the existence of a quantum spin-orbital liquid (QSOL) over a wide parameter
regime around the SU(4)-symmetric point using MFT, LFWT, and VMC",
    "Identifies distinct ordered phases: ferromagnetic state with staggered orbital order (FM AFOxz),
and noncollinear spin dimers (NCD) coupled to a vortex orbital state",
    "Constructs a comprehensive phase diagram as a function of hopping ratio  $\beta = t_2/t$  and
Hund's coupling  $\gamma = JH/U$ "
  ],
  "methodology": "Combination of mean-field theory (MFT) for classical ordered states, linear flavor-
wave theory (LFWT) for quantum fluctuation analysis and stability criteria, and variational Monte
Carlo (VMC) with Gutzwiller projection for studying QSOL and valence-bond-crystal states",
  "datasets_used": [
    "Not specified"
  ],
  "experiments": "VMC calculations performed on honeycomb lattices of linear length  $L = 6, 12,$  and
 $18$  ( $N = 2L^2$  sites) using Monte Carlo moves that exchange random pairs of sites with distinct
colors; energies of FM AFOxz, QSOL, and NCD states compared as functions of  $\beta$  and  $\gamma$ ;
stability of tetramerized and dimerized valence-bond-crystal states tested against the  $c_0$ -flux
QSOL",

```

"results": "The QSOL extends away from the SU(4)-symmetric point and covers an appreciable portion of the phase diagram before giving way to the FM AFOxz phase at $\phi \approx 0.175$ (essentially independent of ϕ). The AFM AFOxz state is unstable against quantum fluctuations (complex dispersion frequencies in LFWT). No competitive valence-bond-crystal candidates were found within VMC. The NCD phase is the best variational state in the region $\phi \approx 0.2$ and $\phi \approx 0.2$. The phase diagram agrees qualitatively with exact diagonalization results from Ref. [31].",

"limitations": [

"The microscopic model (Eq. 5) is too simplistic to fully describe BaCuSbO; a more realistic Hamiltonian on a decorated honeycomb lattice is needed",

"Longer-range exchange couplings, likely relevant in TBG, are not included",

"LFWT cannot diagnose the characteristics of the unstable phase in the gray region of the phase diagram",

"No competitive VMC candidate found for the LFWT-unstable region",

"Results are not directly extendable to systems on triangular lattices such as TLG/hBN"

],

"future_work": [

"Investigation of more realistic Hamiltonians on decorated honeycomb lattices for BaCuSbO",

"Study of trilayer graphene/hBN heterostructures (TLG/hBN) using similar KK models",

"Analysis of tiny-angle TBG under electric fields possibly described by KK models on emergent kagome lattices",

"Exploration of extended models for ZrCl_2 to connect QSOL and NCD phases with its magnetism",

"Investigation of possible spin-polarized phases in proximity to QSOL in twisted bilayer graphene"

],

"relevant_elements": [

"Equation (5): Full Kugel-Khomskii model Hamiltonian $H(\phi, \phi)$ ",

"Equation (8): Explicit spin-orbital decomposition of $H(\phi, \phi)$ ",

"Equation (9): SU(4) Heisenberg model $H_{\text{SU}(4)}$ at the symmetric point",

"Equation (14): LFWT Hamiltonian in Fourier space (12x12 block diagonal form)",

"Equation (17): LFWT ground state energy E_{LFWT} including only Goldstone modes",

"Equation (19): Parton mean-field Hamiltonian for QSOL ansatz",

"Equation (21): Tetramerization order parameter $r_{\text{tet}} = 4/5 (P_1 - P_2)$ ",

"Figure 1: Phase diagram of the KK model as function of ϕ and ϕ ",

"Figure 3: Mean-field phase diagram with classical ordered states",

"Figure 4: LFWT dispersion relations, order parameter corrections, and ground state energies",

"Figure 5: Mean-field ansatz for ϕ -flux QSOL state",

"Figure 6: Energy comparison of FM AFOxz, QSOL, and NCD phases",

"Figure 8: VMC energies of tetramerized and dimerized valence-bond-crystal states",

"Equation (12): Orbital projector form of $H(0, 0)$ at isotropic hopping limits"

],

"visual_analysis_findings": []

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ANALISI LATEX:

comando: python main.py analyze --latex downloaded_latex/"1751397_SU(4) Heisenberg model on the honeycomb lattice wi_source.tar.gz" --model gemini

OUTPUT:

Output JSON:

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{
  "problem_addressed": "The paper investigates the effects of exchange-frustrated perturbations and Hund's coupling on the SU(4) Heisenberg model on the honeycomb lattice, aiming to understand the phases relevant to Mott insulators such as Ba3CuSb2O9 and the insulating phase of magic-angle twisted bilayer graphene at quarter filling.",
  "main_contributions": [
    "Provides a unified theoretical framework for SU(4) Heisenberg model with Hund's coupling and anisotropic, orbital-dependent exchange interactions, applicable to correlated insulators and twistrionic systems.",
    "Shows, using mean-field theory, linear flavor-wave theory, and variational Monte Carlo, that a quantum spin-orbital liquid (QSOL) phase exists over a wide parameter regime around the SU(4)-symmetric point.",
    "Identifies a phase diagram comprising three distinct phases: QSOL, ferromagnetic antiferro-orbital ordered state (FM AFOxz), and a valence-bond crystal combined with a vortex orbital state (noncollinear spin dimers, NCD).",
    "Demonstrates the stability of the QSOL against finite Hund's coupling and orbital anisotropy, and shows its persistence up to a Hund's coupling  $\sim 480.175$ , largely independent of the hopping ratio  $t$ .",
    "Connects the theoretical results to experimental platforms such as Ba3CuSb2O9 and twisted bilayer graphene, discussing possible observations of QSOL and nearby spin-polarized phases."
  ],
  "methodology": "The study combines mean-field theory (MFT) to determine classical ordered states, linear flavor-wave theory (LFWT) to assess quantum fluctuation effects and stability, and variational Monte Carlo (VMC) to compute ground state energies of competing states (QSOL, FM AFOxz, NCD) and construct the phase diagram via energy crossings.",
  "datasets_used": [],
  "experiments": "The authors evaluated the ground state energy per site of the QSOL, FM AFOxz, and NCD states using VMC on honeycomb lattices of linear size  $L=6,12,18$  ( $N=2L^2$  sites) with Gutzwiller-projected parton wavefunctions. They compared these energies to those obtained from LFWT and Huse-Elser variational energies for the FM AFOxz state. Phase boundaries were determined by locating crossing points of the energy curves as functions of Hund's coupling  $J$  for fixed hopping ratio  $t$  (e.g.,  $t=0.2, 0.25$ ). Additionally, MFT yielded a classical phase diagram, and LFWT provided dispersion relations and order parameter corrections to assess stability.",
  "results": "The phase diagram (Fig. \ref{fig:phase_diagram}) shows three phases: a quantum spin-orbital liquid (QSOL, blue) around the SU(4)-symmetric point, a ferromagnetic state with staggered orbital order (FM AFOxz, orange) at large Hund's coupling, and a noncollinear ordering of spin dimers coupled to a vortex orbital state (NCD, green) stabilized by dominant orbital-dependent exchange. The QSOL phase extends to  $J \sim 480.175$  essentially independent of  $t$ , beyond which FM AFOxz appears. The NCD phase appears for  $t \sim 2730.2$  and  $J \sim 22720.2$ . LFWT predicts an instability region (gray) where the FM AFOxz state becomes unstable due to negative mode energies. These findings agree qualitatively with earlier exact diagonalization studies on small clusters."
```

"limitations": "The authors note that the microscopic model is too simplistic to describe real materials like Ba₃CuSb₂O₉, which may require a decorated honeycomb lattice. LFWT cannot diagnose the nature of the unstable region indicated in gray. VMC results are limited to finite cluster sizes, and extrapolation to the thermodynamic limit relies on assumed scaling. The study does not incorporate longer-range exchange couplings that might be relevant in twisted bilayer graphene.",

"future_work": "Future directions include observing the QSOL in honeycomb lattice materials with active spin and orbital degrees of freedom, exploring the proximity of a spin-polarized phase to the QSOL in twisted bilayer graphene, applying the methodology to other platforms such as trilayer graphene/hexagonal boron nitride heterostructures, tiny-angle twisted bilayer graphene, and $j=3/2$ compounds like UO_2ZrCl_3 , and investigating the effects of longer-range exchange interactions and lattice decorations on the phase diagram.",

"relevant_elements": [

"Eq. (\ref{eq:complete_Hamiltonian}): full Kugel-Khomskii model Hamiltonian with exchange-frustrated perturbations and Hund's coupling",

"Eq. (\ref{eq:SU4}): SU(4) Heisenberg model Hamiltonian",

"Eq. (\ref{eq:HI}): onsite interaction Hamiltonian including Hubbard U and Hund's coupling J_H ",

"Eq. (\ref{eq:kinetic}): tight-binding Hamiltonian for hopping on the honeycomb lattice",

"Eq. (\ref{eq:T}): parameterization of the hopping matrix in terms of t and t_{2032} ",

"Mean-field theory (MFT) for classical ordered states",

"Linear flavor-wave theory (LFWT) for quantum fluctuation corrections",

"Variational Monte Carlo (VMC) with Gutzwiller projection for ground state energies",

"Huse-Elser variational wavefunction for the FM AFOxz state",

"Fig. \ref{fig:phase_diagram}: constructed phase diagram",

"Fig. \ref{fig:detail_diagram}: energy comparison of FM AFOxz, QSOL, and NCD",

"Fig. \ref{fig:LFWTdispDS}: flavor-wave dispersions and order parameter correction",

"Fig. \ref{fig:MFTdispersion}: mean-field ansatz and dispersion for the UO_2 -flux state",

"Fig. \ref{fig:Classical_diagram}: classical ordered states (FM/AFM, ferro-/antiferro-orbital, vortex)",

"Fig. \ref{fig:tetra}: tetramerization and dimerization order parameter energies",

"Fig. \ref{fig:trivial_paramagnets}: valence bond crystal states considered"

],

"visual_analysis_findings": []

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----- ANALISI CON PIPELINE VISIVA -----

comando: python main.py full-analyze --pdf downloaded_pdfs/"1751397_SU(4) Heisenberg model on the honeycomb lattice wi.pdf" --model gemini

Output JSON:

{

"problem_addressed": "The paper investigates how Hund's coupling and bond-dependent, orbital-anisotropic exchange interactions modify the SU(4)-symmetric quantum spin-orbital liquid in a Kugel-Khomskii model on the honeycomb lattice, with applications to Ba₃CuSb₂O₉, quarter-filled twisted bilayer graphene, and related materials.",

"main_contributions": [

"Develops a unified analysis of a honeycomb Kugel-Khomskii model incorporating Hund's coupling and exchange frustration from orbital-dependent hopping.",

"Constructs a phase diagram identifying a robust quantum spin-orbital liquid, a noncollinear spin-dimer phase, and a ferromagnetic antiferro-orbital phase.",

"Combines mean-field theory, linear flavor-wave theory, and variational Monte Carlo to assess ordered, liquid, and valence-bond-crystal states.",

"Shows that the SU(4) quantum spin-orbital liquid remains stable over a substantial region around the SU(4)-symmetric point."

],

"methodology": "The authors derive a second-order Kugel-Khomskii Hamiltonian from a two-orbital Hubbard model with onsite interactions and orbital-dependent nearest-neighbor hopping. Mean-field theory evaluates classical product states, while linear flavor-wave theory uses Holstein-Primakoff bosons to calculate dispersions, zero-point energies, and order-parameter reductions. Variational Monte Carlo projects fermionic parton states, including the ϕ -flux quantum spin-orbital liquid and Huse-Elser states, and compares their energies with tetramerized, dimerized, and noncollinear spin-dimer valence-bond-crystal trial states.",

"datasets_used": [],

"experiments": "The model was analyzed over the dimensionless parameters $t=U/2$ and $J=JH/U$ using analytical mean-field and flavor-wave calculations and variational Monte Carlo. The quantum spin-orbital liquid calculations used honeycomb lattices with linear sizes $L=6, 12$, and 18 , approximately 10^5 Monte Carlo sweeps, and half the sweeps discarded for equilibration. Phase boundaries were determined by comparing variational energies, while flavor-wave stability was assessed through complex or negative excitation frequencies.",

"results": "A stable quantum spin-orbital liquid occupies a wide region around $t=J=0$ and survives finite Hund's coupling and orbital anisotropy. It is replaced by a ferromagnetic spin and antiferro-orbital ordered phase at $J \approx 0.175$, essentially independently of t . The noncollinear spin-dimer phase with ferro-orbital vortex order is the best variational candidate for approximately $t \approx 0.265$ and $J \approx 0.264$. The antiferro-orbital antiferromagnetic state is unstable for nonzero t and J . Linear flavor-wave theory gives an order-parameter reduction of approximately 0.14 for $t < 0.99$ and predicts divergence as t approaches 1 . Huse-Elser variational energies agree closely with the flavor-wave energies. Tetramerization and collinear orbital dimerization are not energetically favored, and no competitive variational state was found for the region unstable within flavor-wave theory.",

"limitations": "The minimal nearest-neighbor Kugel-Khomskii model is considered too simplistic to describe $\text{Ba}_3\text{CuSb}_2\text{O}_9$ accurately, and a QSOL is likely not its ground state. The spin-orbital decoupling used for some valence-bond-crystal trial states should fail near the SU(4) point. Longer-range exchange interactions, omitted from the model, are likely important in twisted bilayer graphene. The honeycomb-lattice results cannot be directly extended to triangular-lattice TLG/hBN or kagome-lattice tiny-angle TBG systems. The nature of the flavor-wave-unstable region remains unidentified.",

"future_work": "The authors suggest using more realistic decorated-honeycomb models for $\text{Ba}_3\text{CuSb}_2\text{O}_9$, including longer-range exchanges relevant to twisted bilayer graphene, and applying the methodology to TLG/hBN, tiny-angle TBG on an emergent kagome lattice, and extended Kugel-Khomskii models for ZrCl_3 .",

"relevant_elements": [

"Equations (2)-(4): Two-orbital Hubbard interactions, tight-binding Hamiltonian, and orbital-dependent hopping parameterization.",

"Equation (5): Bond-dependent Kugel-Khomskii Hamiltonian $H(t, J)$, with $J_1=J/(1+J)$, $J_2=J/(1-J)$, $J_3=J/(1+J)$, and $J=t^2/U$.",

"Equation (9): SU(4)-symmetric Heisenberg Hamiltonian $H_{\text{SU}(4)} = 2J \sum_{\langle ij \rangle} (2S_i \cdot S_j + 1/2)(2S_j \cdot S_i + 1/2)$.",

"Equation (14): Quadratic linear-flavor-wave Hamiltonian for the ferromagnetic antiferro-orbital state.",

"Equation (17): Flavor-wave expression for the ground-state energy per site.",

"Equation (18): Fermionic parton representation of SU(4) color-exchange operators.",

"Equation (19): c0 -flux parton mean-field Hamiltonian.",

"Equation (20): Mean-field Hamiltonian for tetramerized trial states.",

"Equation (21): Tetramerization order parameter $r_{\text{tet}} = 4(P_1^2 - P_2^2)/5$.",

"Figure 1: Phase diagram of the QSOL, noncollinear spin-dimer, and ferromagnetic antiferro-orbital phases in the be-b7 plane.",

"Figure 6: Variational energy comparison among the QSOL, ferromagnetic antiferro-orbital state, and noncollinear spin-dimer state.",

"Figure 8: Energy scans showing that tetramerized and collinearly dimerized states are not favored."

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"visual_analysis_findings": []

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LATEX:

comando: python main.py full-analyze --latex downloaded_latex/"1751397_SU(4) Heisenberg model on the honeycomb lattice wi_source.tar.gz" --model gemini

Output JSON:

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  "main_contributions": [
    "Extended Kugel\u2013Khomskii model including Hund\u2019s coupling and anisotropic exchange for the honeycomb lattice.",
    "Phase diagram obtained from a combination of mean\u2011field theory, linear flavor\u2011wave theory and variational Monte\u2011Carlo.",
    "Demonstrated a robust quantum spin\u2011orbital liquid phase around the SU(4)-symmetric point.",
    "Identified a ferromagnetic antiferro\u2011orbital ordered state for large Hund\u2019s coupling.",
    "Found a valence\u2011bond crystal coupled to a vortex orbital state stabilized by dominant orbital\u2011dependent exchange.",
    "Qualitative agreement of the phase diagram with exact\u2011diagonalisation results of Ref.\u202f[Smerald2014].",
  ],
  "methodology": "Mean\u2011field theory (MFT) to locate classical ordered states, linear flavor\u2011wave theory (LFWT) to incorporate quantum fluctuations and assess stability of those states, and variational Monte\u2011Carlo (VMC) with Gutzwiller projection to evaluate ground\u2011state energies of the QSOL, tetramerized, dimerized and non\u2011collinear spin\u2011dimer (NCD) trial wave functions.",
}
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"datasets_used": [],

"experiments": "Numerical evaluations using MFT (classical phase diagram), LFWT (diagonalisation of the flavor\u2011wave Hamiltonian to obtain correction to order parameters and zero\u2011point energy) and VMC calculations on honeycomb lattices with linear sizes $L = 6, 12, 18$ (\u224810\u2075 Monte\u2011Carlo sweeps per configuration). Comparison of the resulting energies allows identification of phase boundaries.",

"results": "A quantum spin\u2011orbital liquid survives for a wide region of $(\text{be}, \text{b7})$ around the $SU(4)$ point; the FM antiferro\u2011orbital (AFOxz) phase appears for $\text{b7} \approx 0.175$ (independent of be); a non\u2011collinear spin\u2011dimer (NCD) phase is favored for $\text{be} \approx 0.2$ and $\text{b7} \approx 0.2$; LFWT predicts an instability region for FM AFOxz near $\text{be} \approx 0.2481$; overall the phase diagram qualitatively matches the exact\u2011diagonalisation results of Ref.\u202f[Smerald2014].",

"limitations": "The KK model is a simplified effective description (e.g. for BaCuSbO it neglects further\u2011neighbor hoppings and a more realistic lattice), the mean\u2011field and LFWT approximations ignore gauge fluctuations, and the VMC analysis is restricted to finite sizes and a limited set of trial wave functions.",

"future_work": "Apply the same framework to more realistic models for BaCuSbO , twisted bilayer graphene, TLG/hBN and ZrCl_3 ; include longer\u2011range exchange couplings and a fully fluctuating gauge field; and explore possible connections between the QSOL/NCD phases and emergent phenomena such as superconductivity or orbital magnetism.",

"relevant_elements": [

"Definition of the four local colour states (Eq.\u202f{eq:colors})",

" $SU(4)$ generators $S_a^b(i) = |a\rangle\langle b|$ (Eq.\u202f{eq:Sab})",

"Kugel\u2013Khomskii Hamiltonian $H(\text{be}, \text{b7})$ (Eqs.\u202f{eq:complete_Hamiltonian} & {eq:complete_Hamiltonian-1})",

" $SU(4)$ Heisenberg model $H_{SU(4)}$ (Eq.\u202f{eq:SU4})",

"Mean\u2011field ansatz for the c0 \u2011flux QSOL, $H_{mf} = -\sum_{ij} t_{ij} c_{i,a}^\dagger c_{j,a}$ (Eq.\u202f{eq:Hmf0})",

"Linear flavor\u2011wave theory Hamiltonian (Eq.\u202f{eq:HLFWT}) and flavor\u2011wave spinor B_k (Eq.\u202f{eq:Bk})",

"Tetramerization order parameter $r_{tet} = (4/5)(P_1 - P_2)$ (Eq.\u202f{eq:rtet})",

"Variational Monte\u2011Carlo procedure with Gutzwiller projection and Metropolis updates (Sec.\u202f{IV})",

"Phase diagrams shown in Fig.\u202f{fig:phase_diagram} and Fig.\u202f{fig:Classical_diagram}"

],

"visual_analysis_findings": [

"AI curve-fit approximation of Figure fig2.png (discrete plot): representation of two distinct orbital configurations \u2013 a linear horizontal arrangement and a vertical two\u2011column arrangement (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig3a.png, curve 'Boundary between FM AFOxz and AFM AFOxz phases': $y = \exp(x_0 \cdot (-2.490917)) \cdot x_0^{1.7329482} \cdot x_0$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig3a.png, curve 'Boundary between FM AFO vortex and AFM AFO vortex phases': $y = 0.1548715/x_0$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig3c.jpeg (discrete plot): discrete coordinates for blue and red arrows illustrating the orbital\u2011dependent bond signs (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig3d.jpeg (discrete plot): six vertex coordinates of the central hexagonal ring (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig3e.jpeg (discrete plot): six vertex coordinates of the central hexagon (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig5b.png, curve 'top orange surface': $y = ((x_0 * x_0 * (x_0 * 0.03459537 - 0.10070558) + 0.45612624) * (-0.0155074) * x_0 / 0.094200715 + 1.0090101) / 0.4345418$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig5b.png, curve 'middle green surface': $y = \exp(0.73393583 + x_0 / ((-2.5203638 / x_0))) - 1.3799803$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig5b.png, curve 'middle dark blue surface': $y = 0.34090707 * \log(x_0^{**2} + 0.044194814)$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig5b.png, curve 'bottom orange surface': $y = \exp(x_0 / ((-1.4802014 / x_0))) - 1 * 1.9728234$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig6b.png, curve 'FM AFOxz': $y = 1.2827935 / (x_0 - 1 * 0.3421155)$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig6b.png, curve 'QSOL': $y = -2.2516923 - (-1) * 0.9079697 / (x_0 - 0.3397743)$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig6b.png, curve 'NCD': $y = -2.4122605 + 1.1685249 / (x_0 - 0.44214976)$ (not derived or stated by the paper's authors).",

"AI curve-fit approximation of Figure fig7a.png (discrete plot): six numbered points marking the vertices of a central regular hexagon (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig7b.png (discrete plot): six numbered points labeling the edges of the central hexagon with orange ellipses highlighting horizontal bonds (coordinates listed in the external analysis).",

"AI curve-fit approximation of Figure fig7c.png (discrete plot): six numbered site vertices forming a central hexagon (coordinates listed in the external analysis)."

]

}